

Business Cycle Transmission from the US to Germany – a Structural Factor Approach

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(first version May 2004, this version November 2005)

Abstract:

This paper investigates the transmission of US macroeconomic shocks to Germany using a large-dimensional structural dynamic factor model. This framework allows us to investigate many transmission channels simultaneously, including "new" channels such as stock markets, foreign direct investment, bank lending and the confidence channel. We find that US shocks affect the US and Germany largely symmetrically. Trade seems to be the most relevant transmission channel. Monetary policy reactions to strong price movements seem to play a role as well. No clear conclusion can be drawn yet on the role of financial markets and the confidence channel. Negative domestic influences apparently more than offset positive US influences in the German economy between 1995 and 2000, but the US recession in 2001 appeared to be the main culprit in the German slump.

Keywords: International business cycles, international transmission channels, dynamic factor models, shock identification

JEL-Classification: F02, F41, C13, C32

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1 Introduction

International business cycle linkages have recently returned to the focus of public interest. This renewed interest has its roots in the worldwide economic downturn in 2001. It has often been claimed that the downturn was caused (at least partly) by economic disturbances in the US (cf. Artis, Galvão and Marcellino (2005), Monfort, Renne, Rüffer and Vitale (2004)). In the second half of the 1990s, the US economy experienced a prolonged phase of extraordinarily large productivity gains due to technological advances. Those productivity gains were expected to be long-lasting. Consequently, global demand was boosted, triggering a rapid and exaggerated boom in international stock markets. The subsequent bursting of the stock market bubble contributed notably to the global economic downturn. Its remarkable strength, speed and synchronicity across industrial countries were apparently unexplainable in terms of trade linkages alone. At the same time, financial markets and confidence measures around the globe were particularly affected during the downturn. This led researchers to examine international business cycle linkages more closely, with a particular focus on the international propagation of US shocks and the relevance of the various international transmission channels.

The present article investigates the transmission of macroeconomic shocks from the US to Germany using the large-scale structural dynamic factor model developed by

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This paper was written while I was visiting the Université Libre de Bruxelles and the Deutsche Bundesbank. It was presented at the ECB workshop “The external dimension of the euro area: trade, capital flows and international macroeconomic linkages”, the APF conference in Frankfurt, the CFS summer school in Eltville and at seminars at the Deutsche Bundesbank, the German Council of Economic Experts and the University of Cologne. I am most grateful to Jörg Breitung, Harald Uhlig, Christian Schumacher and two anonymous referees for very helpful comments and suggestions. Many thanks also go to Jörg Döpke, Olivier de Bandt, Juergen B. Donges, Jacopo Cimadomo, Giovanni Lombardo, Jens Weidmann, Jens Ulbrich, Ralf Fendel, Michele Lenza and Heinz Herrmann for valuable comments. All errors are my own.

This paper represents the author's personal opinions and does not necessarily reflect the views of the Deutsche Bundesbank or its staff.

Giannone, Reichlin and Sala (2002). It addresses three main questions which are relevant for forecasters and policymakers.

First, to what extent do US shocks affect the German economy? Our modeling framework enables us to assess the impact of US shocks not only on German economic activity but also on many other variables such as prices and employment.

Second, what are the role and the relevance of different international transmission channels? Besides the more traditional trade channel, "new" channels, i.e. financial markets and the confidence channel, are considered. From a theoretical point of view, it is unclear whether shocks are spread positively or negatively through all these channels (see the detailed discussion in Kose, Prasad and Terrones (2003b)). There is some theoretical support for a positive international transmission. In terms of trade, for example, higher import demand in one country, for example, will boost exports in other countries (cf. Canova and Dellas (1993)). Also, productivity advances may spread internationally through vertical integration (cf. Kose and Yi (2001), Elliott and Fatás (1996)). In terms of financial integration, it enables agents to diversify their risk by investing in different markets, and financial prices become more highly synchronized through arbitrage (Kose, Prasad and Terrones (2003a,b), Doyle and Faust (2002)). Other theoretical arguments, however, point in favor of a negative international propagation. If trade is accompanied by a larger degree of inter-industrial specialization, linkages should be loosened in the presence of industry-specific shocks (cf. Frankel and Rose (1998)). Greater inter-industrial specialization and negative spillovers can also result from mobile capital being reallocated to economies where it is used most productively (Canova and Marrinan (1998), Kalemli-Ozcan, Sørensen and Yosha (2003), Imbs (2004), Heathcote and Perri (2004)). Given the conflicting theories, the question of how globalization, i.e. real and financial integration as well as confidence linkages, affects the international propagation of macroeconomic shocks has to be answered empirically.

Whereas the shock propagation through trade and financial markets has been investigated extensively in the literature, not much attention has yet been directed to the confidence channel. The confidence channel denotes the international transmission of economic and financial shocks through sentiment and/or expectations that might, for example, be caused by "informational cascades" or "fads" (cf. IMF (2001a), Anderton, di Mauro and Moneta (2004)). It is reflected in a shock transmission which goes beyond or is smaller than the propagation through fundamental channels such as trade and financial markets and has been examined in some detail for emerging markets and for periods of financial stress (cf. Pericoli and Sbracia (2003), Dornbusch, Park and

Claessens (2000)). IMF (2001a) and Anderton, di Mauro and Moneta (2004) have recently emphasized the confidence channel for industrial countries and "normal" cyclical periods. Our analysis seeks to shed light on the role of the different transmission channels.

Third, we consider business cycle linkages between the US and Germany during two specific periods and ask how the boom experienced by the US between 1995 and 2000 and the subsequent US recession in 2001 spread to Germany.

Our paper is related to the empirical literature on cyclical interdependencies between countries. Business cycle linkages between the US and Germany have been examined previously by Canova and Marrinan (1998), Artis, Osborn and Perez (2004), Artis, Galvão and Marcellino (2005) and the German Council of Economic Experts (GCEE, 2001). Those analyses find that shocks to the US business cycle are transmitted significantly and positively to Germany. All these studies employ small-scale VAR models (with constant or time-varying parameters). In addition, the GCEE (2001) use the Deutsche Bundesbank's large-dimensional macroeconomic multi-country model. Numerous studies also investigate business cycle linkages between the US and Europe, cf. IMF (2001b), GCEE (2001), Artis, Osborn and Perez (2004), Dalsgaard, André and Richardson (2001), Artis, Galvão and Marcellino (2005), Adjemian, Darracq Parriès and Smets (2004), De Walque and Wouters (2004), Monfort, Renne, Rüffer and Vitale (2004), Dees, di Mauro, Pesaran and Smith (2005). It goes without saying that it is of particular interest to investigate the transmission to Germany, the largest European economy in terms of GDP and population.

Empirical studies focusing on international transmission channels include Otto, Voss and Willard (2001), Kose, Prasad and Terrones (2003b), Imbs (2004) and Baxter and Kouparitsas (2004). Their work is based on cross-country regression models where some measure of co-movement is regressed on variables covering international transmission channels like trade and financial integration. There is some consensus that a positive relationship exists between two countries' business cycle correlations and trade intensity/financial linkages. IMF (2001a) and Anderton, di Mauro and Moneta (2004) find empirical evidence suggesting that the confidence channel may play a role for the shock transmission between the US and Europe.¹ We will discuss the relevance of the confidence channel by investigating how US shocks affect German

¹ According to the IMF (2001a), international confidence measures are more strongly correlated than international real economic activity measures. In addition, residuals from a regression of business confidence on economic activity measures or leading indicators are found to be significantly correlated across the US and the euro area. Anderton, di Mauro and Moneta (2004) show that US confidence Granger causes euro-area confidence.

confidence measures. In the present framework, it will, however, be difficult to disentangle movements of German confidence which only reflect the international spread of US shocks through fundamental channels from those which are not justified by changes in trade and financial markets.

As already outlined above, the models often employed in analyses focusing on aggregate business cycle linkages range from vector autoregressive (VAR) models², fully structural macroeconometric multi-country models³ or, more recently, fully-fledged international DSGE models.⁴ Among the studies which are mentioned above and which examine business cycle linkages between the US and European countries, only Monfort, Renne, Ruffer and Vitale (2004) use a factor model. Other factor-based analyses of international business cycle linkages with a somewhat different country-coverage include Stock and Watson (2005), Kose, Otrok and Whiteman (2003a,b), Lumsdaine and Prasad (2003), Kose, Prasad and Terrones (2003a), Bayoumi and Helbling (2003), Norrbin and Schlagenhauf (1996), Eickmeier and Breitung (2005) and Eickmeier (2005). However, most of these studies use exact factor models and apply them to a small dataset containing only international output measures.⁵ Most of the above-mentioned factor studies, in addition, do not perform structural analysis, i.e. they do not interpret economic co-movements, but confine themselves to estimating a common cycle.⁶ Finally, note that our paper is also related to other applications of large-scale structural dynamic factor models which have become popular in recent years (see, for example, Giannone, Reichlin and Sala (2002, 2004), Sala (2003), Cimadomo (2003) for monetary policy applications).

The main contribution of the present paper is that it applies a large-dimensional structural dynamic factor model to the topic of international business cycle transmission. To our knowledge, it is the first to do so. Studying international business

² For example, Canova and Marrinan (1998), GCEE (2001), Artis, Osborn and Perez (2003), Dees, di Mauro, Pesaran and Smith (2004).

³ For example, Dalsgaard, André and Richardson (2001), IMF (2001b), GCEE (2001).

⁴ See Adjemian, Darraq Parriès and Smets (2004), De Walque and Wouters (2004).

⁵ There are three exceptions. Bayoumi and Helbling (2003) include 35 series of output, investment, consumption and exports of G7 countries and estimate a large-scale dynamic factor model. Eickmeier (2005) investigates economic comovements in the euro area based on a panel of more than 150 variables from key euro-area economies by means of a dynamic factor model. And Eickmeier and Breitung (2005) employ a factor model to study business cycle synchronization between the euro area and central and east European countries. They use more than 200 variables.

⁶ An exception is, again, Eickmeier and Breitung (2005) who identify structural euro-area shocks and investigate their propagation to new EU member states. Also, some of the studies correlate common international factors with potentially explanatory series, such as US shocks, oil prices, trade variables (Stock and Watson (2005), Kose, Otrok and Whiteman (2003a,b), Monfort, Renne, Ruffer and Vitale (2003), Eickmeier (2005)).

cycles in such a framework has various advantages over VAR models⁷ or structural models. Much information can be exploited in dynamic factor models which should allow us to estimate the common driving forces and their propagation more precisely. Moreover, and particularly favorable in our context, it allows us to assess the responses of many variables to macroeconomic shocks. It can therefore be employed to simultaneously assess the relevance of a large number of transmission channels, including the "new" channels, without needing to know the exact workings of these channels. VAR models and cross-country regression models have to cope with collinearity and scarce-degrees-of-freedom problems. In addition, cross-country regression models need to deal with endogeneity problems. Fully structural models can contain a large number of variables. These models at present, however, do not include all and detailed channels. For example, stock markets, confidence, foreign direct investment (FDI), and bank lending are generally missing. The reason may be a lack of consensus on how to model these "new" channels in a wholly structural framework. These models are found to be not able to fully account for the international output correlations, suggesting that the missing channels are of some importance (IMF (2001), GCEE (2001)). These advantages are accompanied by two drawbacks. The interpretation of the outcome is difficult for such a largely reduced form model. A further disadvantage of our model is that we can assess how trade variables, financial market variables etc. react to the shocks, but we do not know how movements in each of these variables *ceteris paribus* affect economic activity in Germany.

This paper is organized as follows. Section 2 describes the data set. Section 3 presents the model and briefly describes estimation techniques and the identification of shocks. Section 4 characterizes US shocks. Section 5 illustrates the impact of US shocks on German key variables, including variables covering the transmission channels. Section 6 analyzes the contribution of US shocks to economic activity in Germany since the mid-1990s. Section 7 concludes.

2 Data

We rely on a large data set with $N = 296$ variables observed over $T = 112$ quarters from 1975 to 2002. The data set contains domestic real and nominal variables for the US and German economies. In addition, it includes measures of the international

⁷ An exception is the global VAR model recently developed by Pesaran, Schuerman and Weiner (2004) and extended in Dees, di Mauro, Pesaran and Smith (2004). Its estimation involves the estimation of small-scale VAR models only, which are then stacked to yield a global large-scale VAR model and, therefore, does not face the same drawbacks as the usual VAR models.

integration of both countries, namely trade and financial variables and confidence indicators. Notice that we consider trade in goods and services separately. In addition, we are able to disentangle long-term interest rates and stock prices and different categories of capital flows, namely FDI, securities as well as credits. It would certainly be interesting (and possible in the present framework) to consider trade and financial market measures at further disaggregated levels. For instance, the relevance of direct versus third-market effects, or the role of trade of intermediate goods versus final goods, could be investigated. This is, however, beyond the scope of this article, and we leave it for future research. Global influences are finally captured by world commodity prices. Outliers are removed. Integrated series are made stationary through differencing and/or deterministic detrending. All series are standardized to have means of 0 and unit variances. For details on the data we refer to Appendix A and Table 1.

3 The model, estimation and identification of US shocks

The series are collected in the $N \times 1$ vector Y_t . It is assumed that Y_t follows an approximate dynamic factor model (cf. Stock and Watson (1998, 2002), Bai and Ng (2002)) and can be represented as:

$$Y_t = X_t + \Xi_t = CF_t + \Xi_t, \quad (1)$$

where F_t is a $r \times 1$ vector of common dynamic euro-area factors and C is an $N \times r$ matrix of factor loadings. X_t and Ξ_t are $N \times 1$ vectors of common and idiosyncratic components. Typically, $r \ll N$.

The common factors are driven by common shocks which are global shocks and/or idiosyncratic shocks which are propagated to other variables, including international variables. Those factors and shocks are the same for all variables, but reactions to changes in the common factors may differ, the latter being reflected in possibly heterogeneous loadings. The vector Ξ_t , by contrast, contains influences which are specific to individual variables or small groups of variables. The idiosyncratic component Ξ_t thus is driven by idiosyncratic shocks which barely, or do not, spread to other variables. Ξ_t may also reflect measurement error. The idiosyncratic components are allowed to be weakly cross-correlated and weakly serially correlated in the sense of Bai and Ng (2002).

We suppose that F_t has a VAR representation:

$$F_t = \sum_{l=1}^p B_l F_{t-l} + v_t, \quad (2)$$

where the B_l are $r \times r$ coefficient matrices and F_t and v_t are $r \times 1$ vectors. It is further assumed that structural common shocks $\tilde{w}_t$ are linearly related to the orthogonal innovations v_t :

$$\tilde{w}_t = G' v_t. \quad (3)$$

Provided that there are enough identifying restrictions on G , the structural shocks $\tilde{w}_t$ can be recovered from the factor innovations. It is important to notice that we are interested only in a subgroup of structural shocks, namely in US shocks that are spread internationally, which we denote by w_t . The ultimate goal is to assess the transmission of these shocks to the German economy.

For this purpose, we first estimate F_t by applying static principal component analysis to Y_t . The dimension of F_t , r , was estimated to be 8 on the basis of the Bai and Ng (2002) criteria. Notice that eight factors explain 50% of the total variance. Figure 1 shows the time series of the estimated factors. A glimpse of the loadings may help to interpret the factors in economic terms. Table 2 reports the largest ten loadings associated with each of the eight factors. The first factor which explains 29% of the factor space is a real US factor, being most closely related to US employment and industrial production. The second factor accounts for 22% of the factor space, and factor loadings are largest for US CPI inflation and interest rates. Factors three to eight contribute to a much lesser extent to economic comovements. The third factor is most closely related to capacity utilization in Germany, the fourth to interest rates in the US, the fifth to wages, confidence, consumption and employment in Germany, the sixth to unit labor costs in Germany, total reserves in the US and interest rates in Germany, the seventh to the US labor market, and the eighth to the construction industry in Germany. Such an exercise should, however, not be conducted without caution. The factors are only identified up to a rotation and cannot be interpreted as such.⁸ In addition, interpretation is made more difficult by the fact that F_t can be a linear combination between not just the dynamic factors, but also their lags. To address these concerns, in

⁸ With an exception: in the international business cycle literature, the first factor is often interpreted as a common cycle.

the following we identify the structural shocks behind common factors and try to disentangle US shocks that have an impact on the German economy from other common shocks.⁹

We first estimate the innovations v_t by following Giannone, Reichlin and Sala (2002) and fit a VAR(1) model to the estimated vector of factors $\hat{F}_t$. The lag order of the VAR model was estimated with the Schwarz information criterion. The structural US shocks w_t cannot be recovered without imposing identification restrictions. Most studies disentangle spillovers from other common shocks by imposing that the former only have a delayed impact on other countries while the latter affect all countries at the same time (cf. Monfort, Renne, Ruffer and Vitale (2004)). The problem with this approach is that spillovers within a period are attributed to common shocks. Since one of the goals of our article is to assess the propagation of US shocks to Germany, we refrain from restricting the impact *a priori*. Instead, we have chosen to adopt another approach. We focus on the main driving forces of the US economy. We then give them an economic interpretation using identification restrictions with respect to US variables and name them US shocks. The disadvantage of this approach is, of course, that it cannot be assured that those US shocks are shocks that have their origin in the US. In the next section, we will, however, provide some evidence that most of them should. But let us first explain in more detail how we recover these shocks.

Following Uhlig (2003), we extract those shocks that explain as much as possible of the forecast error variance of (the common component of) US GDP, where, as in Uhlig (2003), US GDP is taken as a proxy for US economic activity and the forecast horizon is zero to five years. Principal component analysis is applied to (some function of) the impulse responses of US GDP to v_t . It turns out that two shocks are sufficient to explain the overwhelming forecast error variance share (of the common component) of US GDP, 98%.¹⁰ We therefore only focus on two shocks. The next step involves giving the two shocks a structural interpretation as is done in the SVAR literature. Following Peersman (2005), we use a relatively agnostic identification approach and impose short-run sign restrictions on impulse responses of key US variables to identify a US supply shock and a US demand shock. This prevents us from using zero restrictions which are at odds with some theoretical models (see the discussions in Peersman (2005) and in Canova and de Nicolo (2003)). As in Peersman (2005), we impose the following restrictions. A positive supply shock has non-negative effects on output and non-

⁹ An alternative is to rotate the factors themselves. See, for example, Eickmeier (2005).

¹⁰ This needs to be put in relation to the variance share of US GDP growth explained by the common component, which amounts to 70% (Table 3).

positive effects on prices contemporaneously and during the first four quarters after the shock; the short-term interest rate does not increase on impact. A positive demand shock affects output and prices non-negatively instantaneously and during the first four quarters after the shock; the immediate effect on the short-term interest rate is non-negative. These conditions are consistent with the standard aggregate supply-aggregate demand framework and with more complex structural models such as the DSGE model outlined in Smets and Wouters (2003).¹¹ The vector of impulse-response functions of variable i to the shocks $w_t = (w_{1t} \ w_{2t})'$ at horizon h is given by $\Phi_{ih} = \partial y_{it+h} / \partial w_t$. The median impulse responses and the corresponding 90% bootstrapped confidence bands are shown in Figure 2. For details on the estimation of the model, the identification of the shocks and the construction of the confidence bands see Appendix B.

As already emphasized above, Uhlig's (2003) identification scheme helps to disentangle the main US driving forces that spill over to Germany and other common shocks. Besides this advantage, it has two other advantages. First, it prevents us from determining the number of common structural or dynamic shocks. As already indicated above, the number of shocks is not identical with the dimension of F_t if F_t contains, not only the dynamic factors but also their lags. There exist a number of formal (Breitung and Kretschmer (2004), Bai and Ng (2005)) and informal criteria (Forni, Hallin, Lippi and Reichlin (2000)) to determine the number of structural shocks; however, they sometimes lead to inconclusive results. Second, focusing on a smaller shock space when identifying the shocks involves computational gains. They can be substantial, especially when employing an agnostic identification procedure based on a grid search as is done here.

4 Characterization of the US shocks

The identified supply shock accounts for 87% and the real demand shock captures 10% of the forecast error variance of the common component of US GDP over five years.¹² Although not imposed, the impact of the supply shock on the real US economy is long-lasting. The large explanatory power of this shock for fluctuations of US GDP is consistent with the real business cycle view.¹³ The US demand shock has temporary real

¹¹ In such a model, a productivity shock raises production, lowers marginal cost and thus prices and interest rates, the latter being determined by a Taylor-style monetary policy reaction function. See also Canova and de Nicoló (2003) who sketch theoretical models that are consistent with our restrictions.

¹² In the following, we always refer to a 0 to 5-year forecast horizon and the median response.

¹³ This literature claims that productivity shocks account for the bulk of output variation. On the other hand, we are aware that the contribution of these shocks to business cycle fluctuations crucially depends on the identification scheme (see, for example, Galí (1999), Peersman (2005), Canova and de

effects. Note that private investment increases quite strongly in response to the demand shock, whereas government expenditures do not go up significantly, and private consumption only increases shortly before entering into a prolonged decline. If one supposes that the price effect precedes the capacity effect, the demand shock could therefore be related to investment rather than to private consumption or fiscal policy. The demand shock has a larger impact on prices than the supply shock: the forecast error variance of the common component of CPI inflation explained by the demand shock is 16%, compared to 14% which is explained by the supply shock.

As already pointed out above, there is some evidence suggesting that the two shocks have their origin in the US. We performed the factor analysis and the identification for a data set containing US variables only. The impulse responses of US variables based on this reduced data set and on the bi-national data set look very similar. Let us further anticipate some of our results. We find that the contribution of the shocks to the variance of the forecast error of the common components of world oil prices and other world export prices is small – the US supply shock captures 5% of the forecast error variance of world oil prices and only 15% of other world export prices. The US demand shock explains 8% and 14% respectively. The common component's shares are only 34% and 36% for world oil prices and other world export prices respectively. This indicates that the extracted shocks are not world oil price shocks or other global commodity price shocks which may also drive the common component. It further turns out that the US shocks have larger contemporaneous median effects on US GDP than on German GDP, and the shocks display their maximum impact with some delay in Germany compared to the US. Overall, we therefore carefully conclude that most of the two shocks should stem from the US. This is consistent with Ahmed and Park (1993) and Ahmed, Ickes, Wang and Yoo (1993), who find that country-specific shocks are more important in explaining output fluctuations in OECD countries than external shocks. While even a large and relatively closed economy like the US may be substantially influenced by external shocks during specific crises periods, this is not plausible for a longer period such as the one considered here.

5 Impact on the German economy and transmission channels

Let us now focus on impulse responses and variance decompositions of German economic variables to the two US shocks. According to Figure 2, the US shocks display effects in Germany that are largely symmetric to the effects in the US. The US supply

Nicoló (2003) who all apply distinct identification techniques and find relatively low contributions of productivity shocks).

shock leads to a gradual long-lasting rise in German real GDP of up to 0.3%. The median contribution of this shock to the variance of the forecast error of the common component of German GDP amounts to 9%.¹⁴ The output increase, however, is statistically not significant. Note, in contrast, the significantly positive reaction of labor productivity. Prices respond strongly to the shock. They decline significantly on impact and decrease further to -0.4%, where they remain. Monetary policy seems to play a role in accommodating the US supply shock.

The US demand shock has an immediate impact on German GDP of +0.3%. The impact declines thereafter, becoming insignificant after roughly one year. The corresponding median variance contribution is identical to that of the supply shock (9%). This share is quite large given the relatively low explanatory power of this shock for US economic activity. Investment, consumption, employment, capacity utilization, and productivity also increase. Prices rise gradually, but not significantly. German monetary policy seems to counteract the US demand shock after some delay.

Next, we describe the impact of the US shocks on the variables covering the transmission channels. We focus on real trade variables, exchange rates and the terms of trade, stock prices, long-run interest rates and capital flows, as well as German confidence measures. We address exchange rates within the trade block, although they are determined by capital flows as well. Let us also point out that it is difficult to isolate the confidence channel. We focus on survey-based confidence measures yet bear in mind that financial prices and short-term capital flows also reflect movements in confidence.

A look at the variance shares of the common components of the measures covering German transmission channels yields a first impression of their relevance for business cycle comovements between the US and Germany. Shares are relatively large for German exports and imports (50% and 61%; see Table 3). Percentages are notably smaller for trade in services compared to goods. The common components of the real effective exchange rate and the terms of trade also exhibit high variance shares (72% and 58%). Similarly high values are found for German confidence measures. As regards financial markets, the variance share of German long-term interest rates explained by the common component amounts to 71%, but other variance shares (of capital flows and stock prices) are relatively low (2% to 15%). Let us now turn to impulse responses of these variables and variance decompositions.

¹⁴ The variance share of the common component of German GDP growth is 79%.

- Real trade and relative prices. The US supply shock reduces net exports in Germany. This reduction is based on an immediate significant and temporary decline in German exports of services. In addition, German imports become significantly positive after six quarters. The decline in German net exports is accompanied by a real appreciation of the currency – possibly because German prices decline less than US prices during the first year after the shock – and an improvement in the terms of trade due to strongly declining import prices (not reported here). The US demand shock, by contrast, triggers a significantly positive and permanent response of the trade balance, caused by persistently rising exports. Imports also exhibit a positive response; however, this is transitory and smaller than the response of exports. In contrast to the supply shock, the terms of trade worsen and the German currency strongly and permanently depreciates in real terms. The variance decompositions also show that trade is affected by the US shocks to a non-negligible extent. The supply shock explains 4% of the forecast error variance of exports and 14% of imports, and the demand shock captures 34% of exports and 11% of imports. The former accounts for 7% and 9% of the terms of trade and the real effective exchange rate; the corresponding values with respect to the latter are 29% and 17%.
- Financial markets. The US supply shock leads to a lasting increase in German stock prices, paralleling the reaction of US stock prices. German long-term interest rates decline. Net outflows of FDI decrease. The negative instantaneous reaction of net inflows of credits are, however, difficult to interpret. The US real demand shock does not have a significant effect on US stock prices. Consequently, German stock prices are barely affected; only a small immediate increase of the DAX is observed. Long-term interest rates rise after a delay. Net outflows of credits decrease, and net inflows of credits rise significantly. Variance decompositions underscore the fact that the influence of the supply shock on financial prices is larger than that of the demand shock, explaining 33% compared with 5% of the forecast error variance of the DAX and 22% compared with 9% of the forecast error variance of long-term interest rates.¹⁵
- Confidence. The US supply shock triggers delayed positive responses of German business and consumer confidence. The demand shock affects both measures positively in the short run. They decline thereafter.

¹⁵ The large contributions of the US supply shock to fluctuations of the common component of net total capital and FDI outflows are striking. As pointed out above, the variance shares explained by the common components of these variables are, however, very small.

In summary, trade seems to be most important for the international business cycle transmission. The US supply shock may have increased the US supply of inputs, lowered German import prices and raised German real imports. The US demand shock may have triggered an increase in US import demand and hence German exports. Moreover, trade seems to be influenced by relative prices. Real exchange rates and the terms of trade alter competitiveness, consumer spending power and, ultimately, trade in Germany; they also dampen the transmission of the supply shock and enhance the propagation of the demand shock. The trade channel seems to be more important for spreading the demand shock than for transmitting the supply shock. Monetary policy reacts as expected to the strong price movements after the US supply shock, accommodating the latter. It may display real effects in the medium run, consistent with the transmission lags of monetary policy. It is difficult to draw unambiguous conclusions on the role of financial markets and confidence. Stock price and confidence movements could have enhanced the transmission of the US shocks, especially the US supply shock. Capital flows seem to support shock transmission. However, the picture is somewhat blurred, and the variance shares explained by the common components are small. At present, it is not possible to draw clear conclusions on which categories of capital flows are particularly relevant.

Our results regarding the aggregate impact of US shocks on Germany are roughly consistent with results obtained on the basis of VAR models (cf. GCEE (2001), Artis, Osborn and Perez (2004), Artis, Galvão and Marcellino (2005), Canova and Marrinan (1998)). Comparison is somewhat exacerbated by the fact that these studies focus on aggregate shocks to economic activity and do not distinguish between different structural shocks. The effects found in the present paper are somewhat larger than what is usually found by means of structural macroeconomic multi-country models (cf. Dalsgaard, André and Richardson (2001), IMF (2001b), GCEE (2001)) – which was to be expected, given that not all channels are included in those models. Our findings with respect to the transmission channels are roughly in line with those of Artis, Galvão and Marcellino (2005) and Canova, Ciccarelli and Ortega (2004). The former study finds that exchange rates and monetary policy are most important for the international shock transmission and that capital flows may also be supportive, while the latter indicates a significant role for trade, monetary policy and consumer spending power.

6 Latest US boom and recession - to what extent was Germany affected?

In this section, we investigate to what extent Germany was able to benefit from the boom in the US between 1995 and 2000, and whether the current German slump was caused by the US recession in 2001. We assess the historical decomposition of German economic activity with respect to the shocks derived from our model. For this purpose we compare true series with hypothetical series that would have evolved had they been driven solely by the US shocks. Hypothetical series are generated by setting shocks other than the US supply and demand shocks for all points in time to zero and making use of the estimated matrices of factor loadings and VAR coefficients, as well as the matrices associated with the shock identification.¹⁶

German GDP growth was slightly lower than it would have been had it been driven solely by US shocks between 1995 and 2000. The mean (annualized) growth rate of German GDP was 1.7%, and the corresponding rate for the hypothetical series is 1.9%. Interestingly, it was mainly the US demand shock which spread positively to Germany: German GDP would have grown by 2.3% if it had only been driven by the US demand shock, whereas its growth rate would have amounted to 1.7% if it had only been affected by the US supply shock. Other influences must have held down economic activity in Germany and overcompensated positive US influences. The negative movement of employment and private demand in the second half of the 1990s when compared with the movement if only driven by US shocks is striking and supports the widespread view that domestic influences, like overcapacity in the construction sector after the unification and small productivity gains coupled with relatively high nominal wage increases in the eastern part of Germany, suppressed these variables.

The mean growth rate of German GDP in 2001 was 0.5%, whereas the rate of change of the hypothetical series was lower (0.1%). This finding is consistent with Artis, Galvão and Marcellino (2005) who also attribute most of the 2001 downturn in Germany to a shock to US output growth. Our analysis suggests further that it was the negative US demand shock that contributed substantially to the economic slowdown in Germany:

¹⁶ $F_{1975:1}$ is chosen as a starting value for F_t . Notice that not a single, but 6 rotations of the first two principal component shock point estimates satisfy our identification restrictions. We refer here to the median point estimates of the two structural US shocks and the corresponding rotation matrix. Notice also that we re-added the means (but not deterministic trends) to the hypothetical series. Differences between growth rates discussed in this section and official growth rates published by the statistical offices may arise due to the removal of deterministic trends from our series. Detailed results (with respect to different variables and the two shocks) are not reported in the paper, but are available upon request.

German GDP would not have changed at all if it had been driven by the US demand shock only, whereas its growth rate would have amounted to 2.2% if it had been determined exclusively by the US supply shock.

As regards the transmission channels, hypothetical trade series highlight the large US influence. They move very much in parallel with the true series. Hypothetical financial market series and the corresponding true series, by contrast, do evolve similarly. Interestingly, confidence was not much influenced by US shocks in the mid-1990s, but the lines move much more in parallel since the end of the 1990s. This may simply reflect the stronger business cycle transmission via trade or financial markets by the end of the sample period. However, another interpretation is that the confidence channel has become relevant only in the last few years. This would also indicate that it is too early to expect this channel to show up in existing empirical studies. Our results are finally consistent with IMF (2001a) and Anderton, di Mauro and Moneta (2004) who find an increase in the correlation between US and euro-area confidence measures in recent years.

It should also be noted that our model probably overestimates the US contribution to the German expansion and underestimates the contribution of the US recession to the German slump. The reason is that our linear model cannot account for transmission asymmetry. Studies employing non-linear empirical models find that negative real shocks are transmitted to a larger extent internationally than positive shocks (Artis, Galvão and Marcellino (2005), GCEE (2001), Canova, Ciccarelli and Ortega (2004), Osborn, Perez, Sensier (2005)). These asymmetries can be explained in terms of nominal rigidities (which are stronger downwards than upwards), menu costs, difficulties for firms facing stronger demand to expand their capacities, and informational asymmetry between lenders and borrowers (Ball and Mankiw (1994), Peersman and Smets (2002)).

7 Conclusion

In this paper, we have investigated the transmission of US macroeconomic shocks to the German economy between 1975 and 2002 by means of a large-scale structural dynamic factor model. This framework allows us to simultaneously assess the responses of a large set of real and nominal German variables and investigate the role of many transmission channels, including "new" channels such as stock markets, foreign direct investment, international bank lending and the confidence channel. To that extent, it has

advantages over other models used in this context, which are not able to investigate as many transmission channels simultaneously.

We have identified two US shocks: one supply shock and one real demand shock. We find that these shocks affect the US economy and the German economy largely symmetrically. That is, the supply shock raises output and lowers prices and interest rates, while the demand shock increases all three variables in both countries. Reactions of German output to the US supply shock and of German prices to the US demand shock, however, are not statistically significant.

As concerns the transmission channels: trade, influenced by relative price movements, seems to play the dominant role in the transmission, especially for the propagation of US demand shocks. Besides trade, monetary policy reacts to relatively strong German price movements following US supply shocks and seems to influence the medium-run impact of US shocks. No clear conclusion can be drawn on the role of financial markets and confidence. Interestingly, German confidence has been driven notably by US shocks only since the end of the 1990s. This might indicate that the confidence channel has become relevant only in the last few years.

Historical decompositions, finally, show that negative domestic factors more than compensated for positive US influences during the US boom between 1995 and 2000 in Germany. By contrast, the US recession in 2001 seemed to be the main culprit for the German slump.

Appendix A

The data set incorporates 296 variables. The data are selected such that the US and German economies, as well as the international integration of both countries, are represented in a balanced way. Less data are generally available for Germany than for the US. We thus mainly confine ourselves to including series which are available for both countries. The data set contains variables covering the real side and the nominal side of the US economy and the German economy. In addition, variables approximating the international economic integration of the two countries and those capturing global factors are included, as outlined in the main text. The data are taken from various national and international sources. They are seasonally adjusted and quarterly. This frequency is chosen in order to include national accounts series, which are generally not available on a monthly basis. Originally monthly series were converted into quarterly series. The X11 seasonal adjustment method was applied to originally not seasonally adjusted series. Logarithms are taken for all non-negative series that were not already in rates or percentage units.

The study is carried out for the period from the first quarter of 1975 to the fourth quarter of 2002. One reason for selecting this period is that important capital controls were abolished in Germany in 1974. In the US, the last capital controls were abandoned in 1973. Moreover, this period corresponds to the post-Bretton Woods flexible exchange rate regime period. Another advantage of this starting date is that potentially extraordinary influences of the first oil price shock in 1973-74 are eliminated.

When constructing the data set, one problem to be addressed is the break in the series resulting from German unification. Most German series were extended by applying west German growth rates to the German levels retrospectively from 1991 onwards. Visual inspection of the series does not suggest a break.

Non-stationarity is removed by differencing and/or deterministic detrending. We performed the analysis with several data sets which differ in the treatment of the series. Results are robust across different specifications. A detailed description of the set finally used may be found in Table 1.

Appendix B

This appendix describes the estimation of the factor model, the identification of the structural US shocks and the construction of the confidence bands. We first estimate F_t by applying static principal component analysis to Y_t .

$$\hat{F}_t = \hat{V}' Y_t, \quad (A1)$$

where $\hat{V}$ is the $N \times r$ matrix of eigenvectors corresponding to the largest r eigenvalues of the sample correlation matrix. $\hat{V}$ is an estimate of the matrix of factor loadings C . The estimated vector of static factors $\hat{F}_t$ has a VAR(1) representation, and OLS is applied to each equation yielding the estimated coefficient matrix $\hat{B}$ and the reduced form VAR residuals $\hat{u}_t$.

The residuals $\hat{u}_t$ are orthogonalized, as in Uhlig (2003), by means of the Cholesky decomposition, but any other orthogonalization would work as well. Hence $\text{cov}(\hat{u}_t) = \hat{A}\hat{A}'$, with $\hat{A}$ being the $r \times r$ lower triangular Cholesky matrix. Then

$$\hat{v}_t = \hat{A}^{-1} \hat{u}_t \quad (A2)$$

and $\text{cov}(\hat{v}_t) = I_r$. The vector of impulse response functions of variable y_{it} in period k to v_t is $\varphi_{ik} = c_i B^k A$, where c_i is the i th row of C , and the corresponding variance of the k -step-ahead forecast error is $\varphi_{i0}\varphi'_{i0} + \dots + \varphi_{ik}\varphi'_{ik}$.

We will now attempt to identify the main driving forces of the US economy. We label the vector of these driving forces $\omega_t = (\omega_{1t} \ \omega_{2t} \ \dots \ \omega_{rt})'$, with ω_{jt} , $j = 1, \dots, r$, being scalars, and we suppose that v_t is linearly related to them through the $r \times r$ matrix Q : $v_t = Q\omega_t$. The aim is to choose Q so that the first shock explains as much of the forecast error variance of (the common component of) US GDP over a certain horizon k as possible, the second shock (which is orthogonal to the first) explains as much of the remaining forecast error variance as possible, etc. US GDP is a proxy of economic activity. We choose k to be 19, which yields the variance of the five-years-ahead forecast error.¹⁷ Because v_t is the vector of orthogonal shocks, we can write the forecast error variance accounted for by r shocks as the sum of the forecast error variance accounted for by each shock. Let us now focus on a single or the first shock.

¹⁷ Uhlig (2003) and Altig, Christiano, Eichenbaum and Linde (2002) also choose US output and $k = 19$.

The forecast error variance accounted for by this shock is $\phi_{i0} q_1 q_1' \phi_{i0}' + \dots + \phi_{ik} q_1 q_1' \phi_{ik}'$, where i stands for US GDP and q_1 is the first column of Q . Uhlig (2003) shows that q_1 should be chosen such that $q_1' S_{ik} q_1$ is maximized, where $S_{ik} = (k+1-0)\phi_{i0}' \phi_{i0} + \dots + (k+1-k)\phi_{ik}' \phi_{ik}$, subject to $q_1' q_1 = 1$, which is a normalization condition.¹⁸ The Lagrangian may be set up as follows:

$$L = q_1' S_{ik} q_1 - \gamma(q_1' q_1 - 1), \quad (A3)$$

where γ is the Lagrange multiplier. The solution to this problem q_1 is the first eigenvector of S_{ik} and γ is the corresponding eigenvalue. The shock associated with q_1 may therefore be called the first principal component shock. Hence, Q is the matrix of eigenvectors of S_{ik} , $(q_1 \ q_2 \ \dots \ q_r)$, where q_j is the eigenvector corresponding to the j^{th} principal component shock and $j = 1, \dots, r$.

As explained in the main text, two shocks explain the bulk, i.e. 98%, of the variance of the forecast error of the common component of US GDP, given by the sum of the first two eigenvalues of S_{ik} . Hence, we concentrate on the first two estimated principal component shocks, represented by $\omega_t' = (\hat{\omega}_{1t} \ \hat{\omega}_{2t})'$, and neglect the remaining ones.

Up to now, the principal component shocks are identified up to a rotation. The vector of principal component shocks ω_t is linearly related to the vector of structural shocks $w_t' = (w_{1t} \ w_{2t})'$ through the 2×2 orthonormal rotation matrix R :

$$\hat{w}_t = R' \hat{\omega}_t, \quad (A4)$$

with $\text{cov}(\hat{w}_t) = I_2$ and $R'R = I_2$. The objective is now to fix a rotation which yields plausible results in terms of impulse-response functions and variance decompositions. Suppose that the first two columns of AQ correspond to the $r \times 2$ -dimensional impulse matrix associated with the first and the second principal component shocks, called Λ . Then the 2×1 vector of the impulse response functions of variable i to the shocks w_t at horizon k can be computed¹⁹ as $\psi_{ik} = c_i B^k \Lambda R$ and the corresponding forecast error variance as $\psi_{i0} \psi_{i0}' + \dots + \psi_{ik} \psi_{ik}'$.

R has to be chosen such that the identifying restrictions specified in the main text are satisfied. Any two-dimensional rotation matrix R can be parametrized as follows

¹⁸ For a detailed derivation see Uhlig (2003).

¹⁹ Impulse responses are also multiplied by the variables' standard deviations.

$$R = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}. \quad (\text{A5})$$

To systematically explore the factor space, the rotation angle θ is varied on a grid from 0 to $\pi/2$. Further rotations would only result in repetitions, possibly with a flipped sign. The number of grids is set at 12, and θ is fixed such that the imposed restrictions are satisfied.

As outlined in the main text, we identified an aggregate supply and an aggregate real demand shock. We also tried to identify a monetary policy shock, where, as in Peersman (2005), we restricted output and prices not to fall instantaneously and during the first four quarters after a positive monetary policy shock and short-term interest rates not to rise contemporaneously. By rotating the first two principal component shocks, we tried to identify such a monetary policy shock together with a supply shock and a monetary policy shock together with a demand shock. In addition, we rotated the first three principal component shocks and tried to simultaneously identify a supply shock, a demand shock and a monetary policy shock. But no rotation which satisfied the respective restrictions at the same time was found, even after increasing the number of grids. This suggests no monetary policy shock among the most important drivers of US economic real activity and is consistent with the view that monetary policy shocks contribute little to business cycles movements (cf. Uhlig (2003)).

Since $N \gg T$, the uncertainty involved with the factor estimation can be neglected (see cf. Bernanke and Boivin (2003)). In order to account for the uncertainty involved in the estimation of the VAR model on the factors, we construct confidence bands by means of the bootstrap-after-bootstrap techniques based on Kilian (1998). These techniques allow us to remove a possible bias in the VAR coefficients which can arise due to the small sample size of the VAR model (for details on the bootstrap see Kilian (1998)). Most draws deliver not just one, but a set of shocks which all satisfy the restrictions. In this case, we follow Peersman (2005) and draw and save one of them. Some draws, however, do not deliver any shocks satisfying the restrictions. We draw until we have saved 500 shocks (536 draws were needed). For more details on the identification, the reader is referred to Peersman (2005).

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Table 1: Data

#	Variables	Treatment ¹⁾	Source ²⁾
US variables			
1	Gross domestic product, real	2	BEA
2	Personal consumption expenditures, real	2	BEA
3	Durables, total	2	BEA
4	Durables, motor vehicles and parts	2	BEA
5	Durables, furniture and household equipment	2	BEA
6	Durables, other	2	BEA
7	Nondurables, total	2	BEA
8	Food	2	BEA
9	Clothing shoes	2	BEA
10	Gasoline fuel oil and other energy goods	2	BEA
11	Other	2	BEA
12	Services, total	2	BEA
13	Housing	2	BEA
14	Housing operation	2	BEA
15	Transportation	2	BEA
16	Medical care	2	BEA
17	Other	2	BEA
18	Gross private domestic investment, real	2	BEA
19	Private fixed investment	2	BEA
20	Nonresidential	2	BEA
21	Nonresidential, structures	2	BEA
22	Nonresidential buildings incl. farms	2	BEA
23	Utilities	2	BEA
24	Mining exploration, shafts, and wells	2	BEA
25	Other structures	2	BEA
26	Nonresidential, equipment and software	2	BEA
27	Information processing and related equipment	2	BEA
28	Industrial equipment	2	BEA
29	Transportation equipment	2	BEA
30	Residential	2	BEA
31	Structures	2	BEA
32	Equipment	2	BEA
33	Change in priv. inventories, real	0	BEA
34	Government consumption expenditures and gross investment, real	2	BEA
35	Employment, total private	2	BLS
36	Goods-producing	2	BLS
37	Natural resources and mining	2	BLS
38	Construction	2	BLS
39	Manufacturing	2	BLS
40	Service-producing	2	BLS
41	Private service producing	2	BLS
42	Trade, transportation and utilities	2	BLS
43	Wholesale trade	2	BLS
44	Retail trade	2	BLS
45	Transportation and utilities	2	BLS
46	Information	2	BLS
47	Financial activities	2	BLS
48	Professional and business services	2	BLS
49	Education and health services	2	BLS
50	Leisure and hospitality	2	BLS
51	Government	2	BLS
52	Unemployed, civilian labor force	1	BLS
53	Unemployment rate	0	BLS
54	Average weekly hours, manufacturing	2	BLS
55	Hours worked, nonfarm business sector	2	BLS
56	Industrial production, total	2	FRB
57	Manufacturing	2	FRB

58	Consumer goods, total	2	FRB
59	Durables	2	FRB
60	Nondurables	2	FRB
61	Business equipment	2	FRB
62	Defense and space equipment	2	FRB
63	Construction supplies	2	FRB
64	Other business supplies	2	FRB
65	Materials	2	FRB
66	Capacity utilization, total	0	FRB
67	Manufacturing	0	FRB
68	Durables	0	FRB
69	Nondurables	0	FRB
70	Other	0	FRB
71	Mining	3	FRB
72	Electric and gas utilities	0	FRB
73	Computers, communications equipment, semiconductors	0	FRB
74	Primary and semifinished processes	3	FRB
75	Finished processing	3	FRB
76	Productivity (output per hour, all persons), nonfarm business sector	2	BLS
77	Unit labor costs, all persons, nonfarm business sector	2	BLS
78	Unit nonlabor payments, all persons, nonfarm business sector	2	BLS
79	Wages and salaries, total, nominal	2	BEA
80	Private industries	2	BEA
81	Commodities-producing industries	2	BEA
82	Manufacturing	2	BEA
83	Distributive industries	2	BEA
84	Service industries	2	BEA
85	Government	2	BEA
86	Disposable personal income, real	2	BEA
87	Personal savings, nominal	2	BEA
88	Saving rate	0	BEA
89	CPI, total	2	BLS
90	Food and beverages	2	BLS
91	Housing	2	BLS
92	Apparel	2	BLS
93	Transportation	2	BLS
94	Medical care	2	BLS
95	Other goods and services	2	BLS
96	Commodities	2	BLS
97	PPI, total (finished goods)	2	BLS
98	Capital equipment	2	BLS
99	Intermediate materials and supplies	2	BLS
100	Materials and components for construction	2	BLS
101	Processed fuels and lubricants	2	BLS
102	Crude materials	2	BLS
103	Implicit price deflator, GDP	2	BEA
104	Personal consumption expenditures	2	BEA
105	Private fixed investment	2	BEA
106	Government consumption and investment expenditure	2	BEA
107	Private inventories	2	BEA
108	M1	2	FRB
109	M2	2	FRB
110	M3	2	FRB
111	Consumer installment loan: total outstanding	2	FRB
112	Real balances (M1/GDP deflator)	2	FRB/BEA
113	Total reserves	2	FRB
114	Non borrowed reserves	2	FRB
115	Consumer confidence, total	3	Conf. Board
116	Consumer confidence, current situation	3	Conf. Board
117	Consumer confidence, expectations	0	Conf. Board
118	Business confidence (ISM Purchasing Managers Index, Manuf. Survey)	0	ISM
119	Business confid. expect. (FRB Philad., 6 months forec., diff. index)	0	FRB Philad.
120	Federal Funds Rate	0	FRB

121	Commercial paper 3 months (non financial)	0	FRB
122	Commercial paper 3 months (financial)	0	FRB
123	CDs secondary market 3 months	0	FRB
124	Treasury bills, constant maturity, 1 year	0	FRB
125	Treasury bills, constant maturity, 10 year	0	FRB
126	Conventional mortgage rates 30 years	0	FRB
127	Moody's AAA seasoned	0	FRB
128	Moody's BBB seasoned	0	FRB
129	Interest rate spread, 3m CDs - Federal Funds Rate Funds Rate	0	FRB
130	Interest rate spread, 1y treasury yield - Federal Funds Rate Funds Rate	0	FRB
131	Interest rate spread, 10y treasury yield - Federal Funds Rate Funds Rate	0	FRB
132	Interest rate spread, Moody's AAA - Federal Funds Rate Funds Rate	0	FRB
133	Interest rate spread, Moody's BBB - Federal Funds Rate Funds Rate	0	FRB
134	S&P 500 Composite, Index	2	Datastream
135	S&P Price Earnings Ratio	0	Datastream
136	Exports total, real	2	BEA
137	Exports goods	2	BEA
138	Exports services	2	BEA
139	Imports total, real	2	BEA
140	Imports goods	2	BEA
141	Imports services	2	BEA
142	Price exports total	2	BEA
143	Price exports goods	2	BEA
144	Price exports services	2	BEA
145	Price imports total	2	BEA
146	Price imports goods	2	BEA
147	Price imports services	2	BEA
148	Terms of Trade (price of exports/price of imports)	2	BEA
149	Net capital outflows, total, nominal	3	BEA
150	Direct investment	3	BEA
151	Securities	3	BEA
152	Credits banks	3	BEA
153	Credits non-banks	3	BEA
154	Net capital inflows, total, nominal	3	BEA
155	Direct investment	3	BEA
156	Securities	3	BEA
157	Credits banks	3	BEA
158	Credits non-banks	3	BEA
159	Exchange rate (US Broad Index), real	2	FRB
160	Current account balance	3	BEA
GER variables			
161	Gross domestic product, real	2	StaBu
162	Consumption of private households, real	2	StaBu
163	Food beverages and tobacco	2	StaBu
164	Clothing and footwear	2	StaBu
165	Housing water and energy	2	StaBu
166	Furnishings and households equipment	2	StaBu
167	Transport and communications	2	StaBu
168	Recreation and culture	2	StaBu
169	Hotel and restaurant services	2	StaBu
170	Other purposes	2	StaBu
171	Private gross fixed capital formation, real	2	StaBu
172	Machinery and equipment	2	StaBu
173	Metal products and machinery	2	StaBu
174	Transport equipment	2	StaBu
175	Construction	2	StaBu
176	Housing	2	StaBu
177	Other construction	2	StaBu
178	Building construction	2	StaBu
179	Civil and underground engineering	2	StaBu
180	Other products	2	StaBu

181	Change in private and public inventories, real	0	StaBu
182	Government consumption expenditures, real	2	StaBu
183	Government fixed investment, real	2	StaBu
184	Employment, total	2	StaBu
185	Agriculture, forestry and fishing	2	StaBu
186	Industry, incl. energy, excl. construction	2	StaBu
187	Construction	2	StaBu
188	Trade and transport	2	StaBu
189	Financial renting and business activities	2	StaBu
190	Other service activities	2	StaBu
191	Unemployment (registered unemployed)	2	OECD MEI
192	Unemployment rate	0	Buba
193	Industrial production, total	2	Buba
194	Manufacturing	2	Buba
195	Construction	2	Buba
196	Chemicals	2	Buba
197	Investment goods	2	Buba
198	Consumer goods	2	Buba
199	Intermediate goods	2	Buba
200	Capacity utilization, manufacturing	0	IFO
201	Manufacturing trade excl. foodstuffs	0	IFO
202	Capital goods production	0	IFO
203	Consumer goods	3	IFO
204	Durable goods	3	IFO
205	Nondurable goods	0	IFO
206	Intermediate goods excl. chemicals	0	IFO
207	Raw materials production excl. chemicals	0	IFO
208	Productivity (output per hour), industry	2	Buba
209	Unit labor costs, total	2	StaBu
210	Wages and salaries, total	2	StaBu
211	Agriculture, forestry and fishing	2	StaBu
212	Industry, incl. energy, excl. construction	2	StaBu
213	Construction	2	StaBu
214	Trade and transport	2	StaBu
215	Financial renting and business activities	2	StaBu
216	Other service activities	2	StaBu
217	Disposable income	2	StaBu
218	Personal savings	2	StaBu
219	Savings ratio	0	StaBu
220	CPI, total	2	OECD MEI
221	Non food, non energy	2	OECD MEI
222	Energy	2	OECD MEI
223	Food and alcohol free drinks	2	OECD MEI
224	Housing rental services	2	OECD MEI
225	PPI, total	2	OECD MEI
226	Investment goods	2	OECD MEI
227	Manufacturing industry	2	OECD MEI
228	Deflator, GDP	2	StaBu
229	Consumption (private and public)	2	StaBu
230	Gross investment (private and public)	2	StaBu
231	Gross fixed investment (private and public)	2	StaBu
232	M1	2	Buba
233	M2	2	Buba
234	M3	2	Buba
235	Loans, lending to domestic enterprises and households	2	Buba
236	Lending to domestic enterprises	2	Buba
237	Lending to domestic self-employed households	2	Buba
238	Lending to domestic employees and other individuals	2	Buba
239	Real balances (M1/GDP deflator)	2	Buba/StaBu
240	Consumer confidence	0	Europ. Comm.

241	Business confidence	0	IFO
242	Business confidence, expectations (next 6 months)	0	IFO
243	Overnight rate	0	Buba
244	3m money market rate	0	Buba
245	Long-term government bonds	0	Buba
246	Mortgage rates 3 years	0	Buba
247	Mortgage rates 10 years	0	Buba
248	Interest rate spread (3m - overnight rate)	0	Buba
249	Interest rate spread (10y - overnight rate)	0	Buba
250	DAX	2	Buba
251	Exports total, real	2	StaBu
252	Exports goods	2	StaBu
253	Exports services	2	StaBu
254	Imports total, real	2	StaBu
255	Imports goods	2	StaBu
256	Imports services	2	StaBu
257	Price exports total	2	StaBu
258	Price exports goods	2	StaBu
259	Price exports services	2	StaBu
260	Price imports total	2	StaBu
261	Price imports goods	2	StaBu
262	Price imports services	2	StaBu
263	Terms of Trade (price of exports/price of imports)	2	Buba
264	Exports total to the US, nominal	2	Buba
265	Exports total to the EU, nominal	2	Buba
266	Exports total to other countries, nominal	2	Buba
267	Imports total from the US, nominal	2	Buba
268	Imports total from the EU, nominal	2	Buba
269	Imports total from other countries, nominal	2	Buba
270	Net capital outflows, total, nominal	3	Buba
271	Direct investment	3	Buba
272	Equity capital	3	Buba
273	Reinvested earnings	3	Buba
274	Intercompany debt	3	Buba
275	Securities	3	Buba
276	Stocks	3	Buba
277	Bonds	3	Buba
278	Credits banks	3	Buba
279	Credits non-banks	3	Buba
280	Net capital inflows, total, nominal	3	Buba
281	Direct investment	3	Buba
282	Equity capital	3	Buba
283	Reinvested earnings	3	Buba
284	Intercompany debt	3	Buba
285	Securities	3	Buba
286	Stocks	3	Buba
287	Bonds	3	Buba
288	Credits banks	3	Buba
289	Credits non-banks	3	Buba
290	Real effective exchange rate DM (Euro)	2	Buba
291	Current account balance	3	Buba
Global variables			
292	Exchange rate US dollar/DM (Euro), nominal	2	Buba
293	World crude petroleum price	2	Datastream
294	World price all exports excl. fuel	2	Datastream
295	World gold price	2	Datastream
296	World food price	2	Datastream

¹⁾ 0: no transformation, 1: logarithm, 2: first difference of logarithm, 3: first difference.

²⁾ BEA: Bureau of Economic Analysis. BLS: Bureau of Labor Statistics. FRB: Federal Reserve Board. FRB Phil.: Federal Reserve Bank of Philadelphia. Conf. Board: Conference Board. ISM: Institute for Supply Management. StaBu: Federal Statistical Office Germany. Buba: Deutsche Bundesbank. IFO: IFO Institute Munich. OECD MEI: OECD Main Economic Indicators.

Table 2: Largest ten factor loadings and variance explained by each factor

Factor 1		Factor 2		Factor 3		Factor 4		Factor 5		Factor 6		Factor 7		Factor 8	
#	Loading	#	Loading	#	Loading	#	Loading	#	Loading	#	Loading	#	Loading	#	Loading
35	0.142	89	0.141	201	0.147	127	0.154	213	0.149	209	0.172	85	0.173	195	0.174
36	0.140	133	-0.136	219	0.145	125	0.152	241	-0.146	113	0.157	47	0.162	179	0.171
41	0.135	227	0.130	200	0.145	128	0.150	162	0.134	249	-0.155	62	0.150	177	0.169
39	0.135	96	0.130	202	0.138	126	0.146	186	-0.134	243	0.136	79	0.144	70	0.152
42	0.133	132	-0.129	240	0.137	161	0.135	176	0.133	237	0.135	84	0.140	175	0.152
118	0.132	104	0.128	292	0.129	246	0.127	175	0.132	108	0.131	219	-0.137	165	-0.147
40	0.131	99	0.126	159	-0.127	105	-0.125	166	0.130	244	0.130	162	0.137	178	0.142
55	0.131	225	0.125	206	0.127	245	0.123	179	0.119	254	-0.129	187	-0.133	134	-0.140
56	0.129	97	0.125	290	0.125	100	-0.121	177	0.116	255	-0.127	70	0.131	291	0.136
57	0.128	131	-0.124	205	0.123	247	0.117	78	0.116	205	0.122	169	0.131	176	0.127
Variance of the total panel explained by each factor/variance explained by the first eight factors															
0.29		0.22		0.12		0.09		0.08		0.07		0.07		0.05	

¹⁾ The variable description corresponding to # can be found in Table 1. The signs of the factors are normalized such that largest absolute loading is positive.

Table 3: Variance decompositions

	Variance shares of the common components ¹⁾ (CC) of		Forecast error variance ¹⁾ of the CC of US variables explained by the		Forecast error variance ¹⁾ of the CC of GER variables explained by the	
	US variables	GER variables	US supply shock	US demand shock	US supply shock	US demand shock
Gross domestic product ²⁾	0.70	0.79	0.87 (0.72 0.94)	0.10 (0.05 0.25)	0.09 (0.01 0.37)	0.09 (0.01 0.31)
Private investment ²⁾	0.79	0.77	0.76 (0.58 0.87)	0.12 (0.05 0.28)	0.08 (0.01 0.34)	0.05 (0.01 0.24)
Personal consumption expenditure ²⁾	0.59	0.81	0.61 (0.35 0.82)	0.17 (0.04 0.41)	0.04 (0.01 0.22)	0.19 (0.02 0.48)
Employment	0.84	0.73	0.77 (0.57 0.88)	0.10 (0.05 0.26)	0.11 (0.02 0.36)	0.22 (0.03 0.56)
Productivity	0.36	0.37	0.17 (0.06 0.42)	0.24 (0.06 0.49)	0.11 (0.02 0.34)	0.08 (0.03 0.26)
Capacity utilization	0.85	0.87	0.28 (0.14 0.47)	0.31 (0.17 0.44)	0.28 (0.15 0.42)	0.25 (0.13 0.42)
Government expenditures ²⁾	0.17	0.50	0.03 (0.00 0.23)	0.06 (0.00 0.30)	0.03 (0.00 0.20)	0.02 (0.00 0.14)
Consumer confidence	0.36	0.64	0.54 (0.30 0.73)	0.13 (0.04 0.36)	0.19 (0.07 0.41)	0.13 (0.04 0.31)
Business confidence	0.81	0.82	0.37 (0.24 0.50)	0.42 (0.29 0.55)	0.20 (0.09 0.35)	0.20 (0.09 0.35)
Consumer prices	0.89	0.76	0.14 (0.02 0.44)	0.16 (0.02 0.46)	0.33 (0.07 0.61)	0.03 (0.00 0.17)
Short-term interest rates ³⁾	0.91	0.86	0.24 (0.08 0.48)	0.36 (0.15 0.60)	0.31 (0.10 0.54)	0.18 (0.07 0.34)
Long-term interest rates ³⁾	0.88	0.71	0.12 (0.02 0.38)	0.21 (0.04 0.47)	0.22 (0.04 0.49)	0.09 (0.02 0.24)
M1 ³⁾	0.61	0.37	0.04 (0.00 0.24)	0.05 (0.01 0.25)	0.16 (0.01 0.48)	0.40 (0.12 0.71)
Stock prices	0.21	0.15	0.27 (0.05 0.55)	0.03 (0.00 0.15)	0.33 (0.06 0.62)	0.05 (0.01 0.24)
Wages ³⁾	0.71	0.44	0.49 (0.18 0.73)	0.17 (0.05 0.45)	0.09 (0.01 0.37)	0.24 (0.03 0.58)
Exports ²⁾ total	0.38	0.50	0.32 (0.06 0.57)	0.10 (0.04 0.35)	0.04 (0.00 0.17)	0.34 (0.09 0.61)
Goods	0.36	0.50	0.33 (0.08 0.58)	0.11 (0.04 0.39)	0.04 (0.00 0.20)	0.33 (0.08 0.61)
Services	0.08	0.13	0.19 (0.02 0.47)	0.05 (0.01 0.23)	0.10 (0.01 0.37)	0.32 (0.09 0.58)
Imports ²⁾ total	0.59	0.61	0.48 (0.24 0.67)	0.10 (0.04 0.24)	0.14 (0.01 0.42)	0.11 (0.02 0.38)
Goods	0.54	0.57	0.47 (0.23 0.67)	0.10 (0.04 0.27)	0.17 (0.01 0.44)	0.11 (0.02 0.38)
Services	0.28	0.17	0.29 (0.04 0.60)	0.20 (0.04 0.52)	0.05 (0.01 0.22)	0.07 (0.01 0.31)
Terms of trade	0.57	0.58	0.08 (0.02 0.25)	0.06 (0.01 0.27)	0.07 (0.01 0.34)	0.29 (0.08 0.55)
Real effective exchange rate	0.66	0.72	0.08 (0.00 0.39)	0.19 (0.01 0.49)	0.09 (0.00 0.40)	0.17 (0.01 0.46)
Net capital outflows, total	0.04	0.03	0.03 (0.01 0.26)	0.11 (0.01 0.36)	0.47 (0.22 0.69)	0.09 (0.03 0.28)
Foreign direct investment	0.03	0.02	0.20 (0.03 0.49)	0.06 (0.01 0.27)	0.23 (0.03 0.53)	0.06 (0.01 0.26)
Securities	0.07	0.05	0.06 (0.00 0.33)	0.28 (0.04 0.57)	0.07 (0.00 0.32)	0.04 (0.00 0.23)
Credits	0.06	0.06	0.06 (0.01 0.33)	0.05 (0.00 0.22)	0.12 (0.01 0.37)	0.15 (0.05 0.38)
Net capital inflows, total	0.03	0.05	0.05 (0.01 0.26)	0.16 (0.01 0.46)	0.09 (0.01 0.33)	0.07 (0.02 0.28)
Foreign direct investment	0.02	0.07	0.13 (0.01 0.37)	0.07 (0.01 0.28)	0.04 (0.00 0.28)	0.12 (0.01 0.46)
Securities	0.02	0.14	0.04 (0.00 0.24)	0.24 (0.01 0.56)	0.04 (0.00 0.18)	0.03 (0.00 0.16)
Credits	0.04	0.02	0.04 (0.01 0.16)	0.05 (0.01 0.23)	0.09 (0.01 0.29)	0.11 (0.04 0.35)

¹⁾ Refers to forecast horizons 0 to 5 years and to the levels of the series. For details on the construction of the (90%) confidence bands see Appendix B.

²⁾ Real

³⁾ Nominal

⁴⁾ Refers to $I(0)$ series, that is, GDP refers to the first difference of GDP etc.

Figure 1: Times series of the first eight common factors

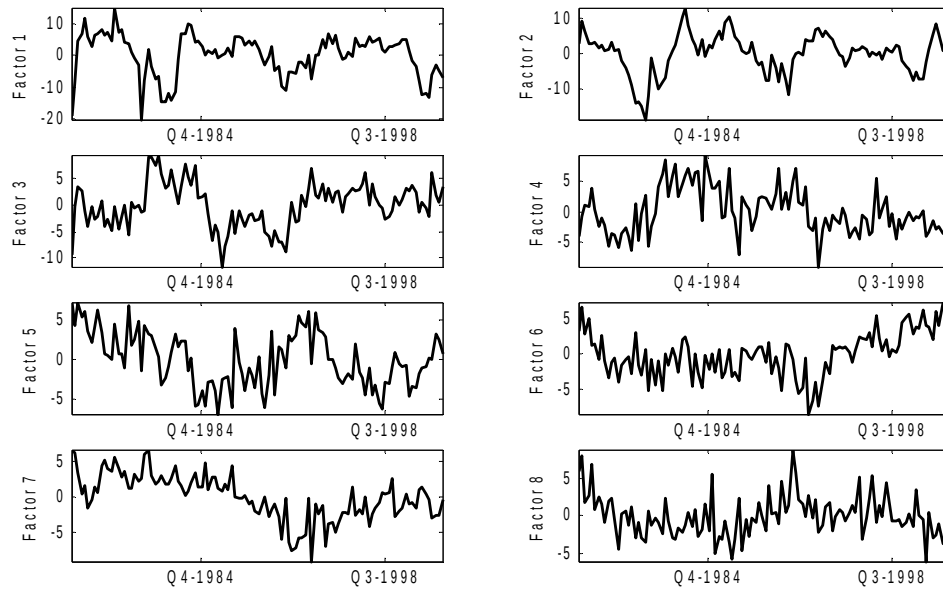
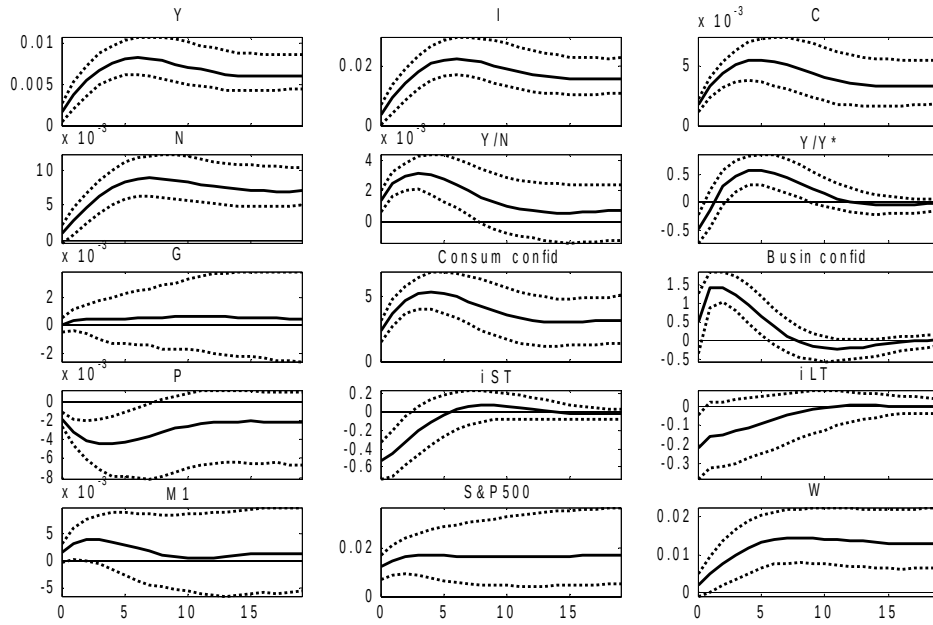
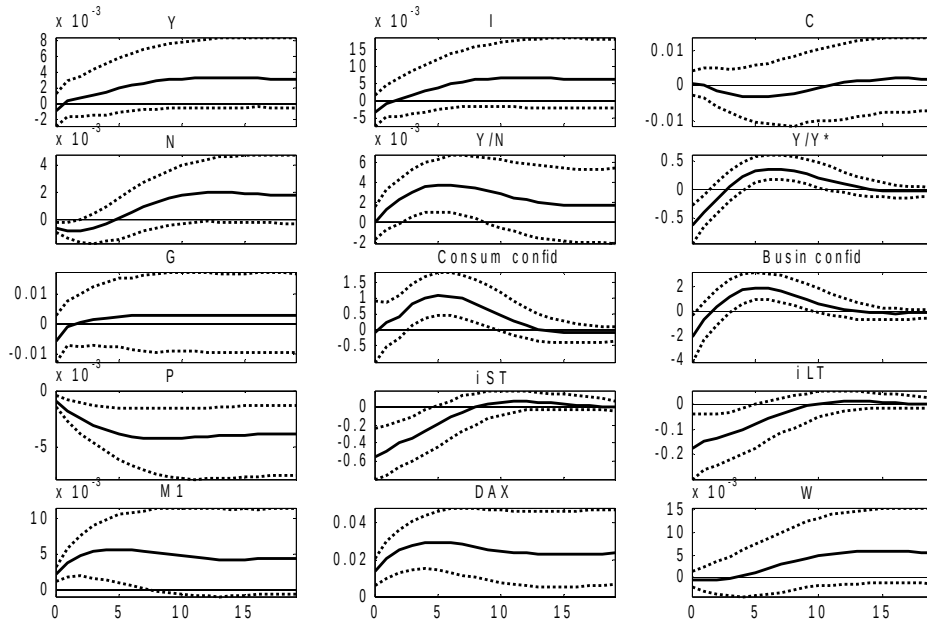


Figure 2: Impulse-response functions to US shocks¹⁾

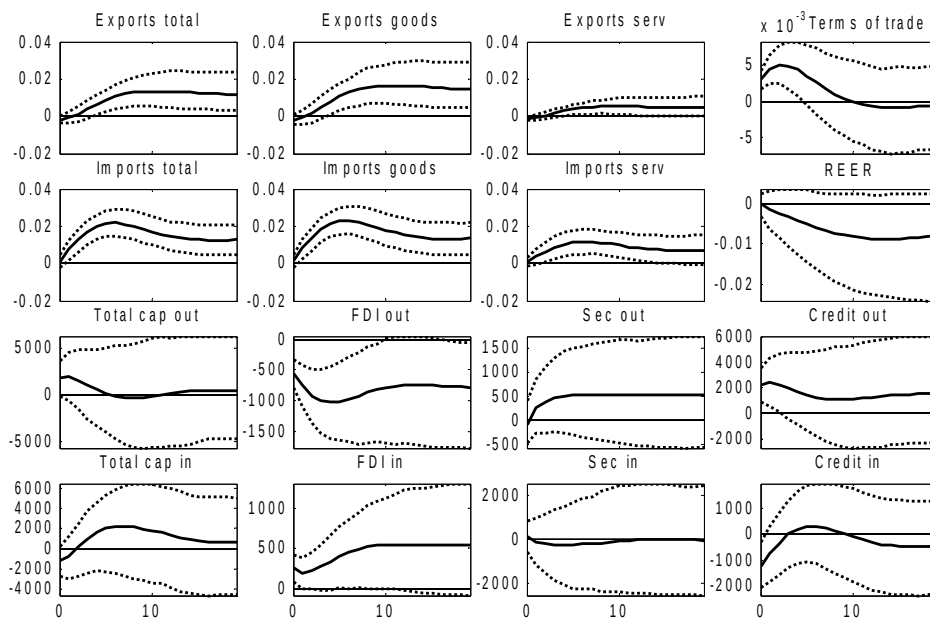
a) US supply shock → US variables



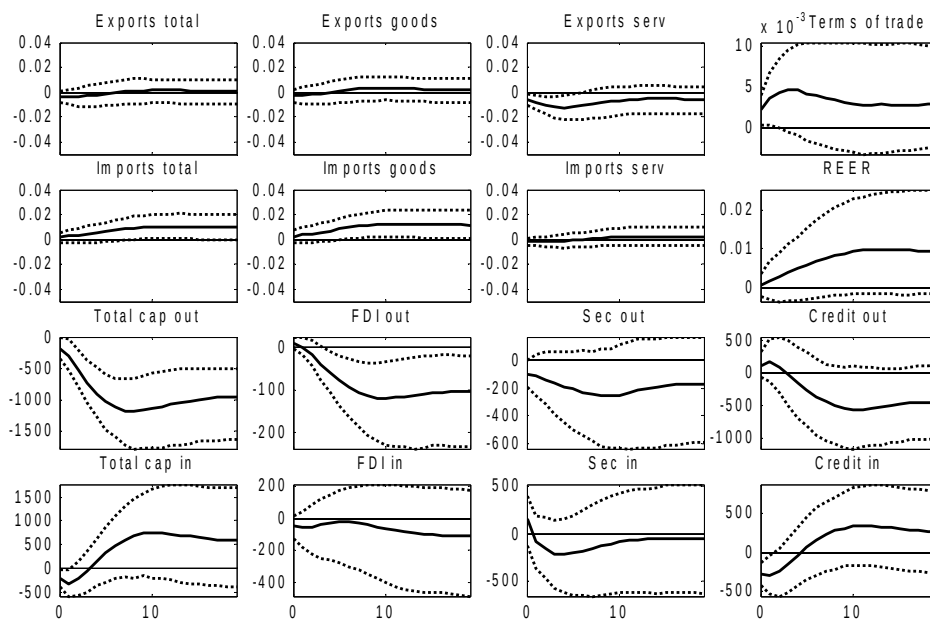
b) US supply shock → German variables



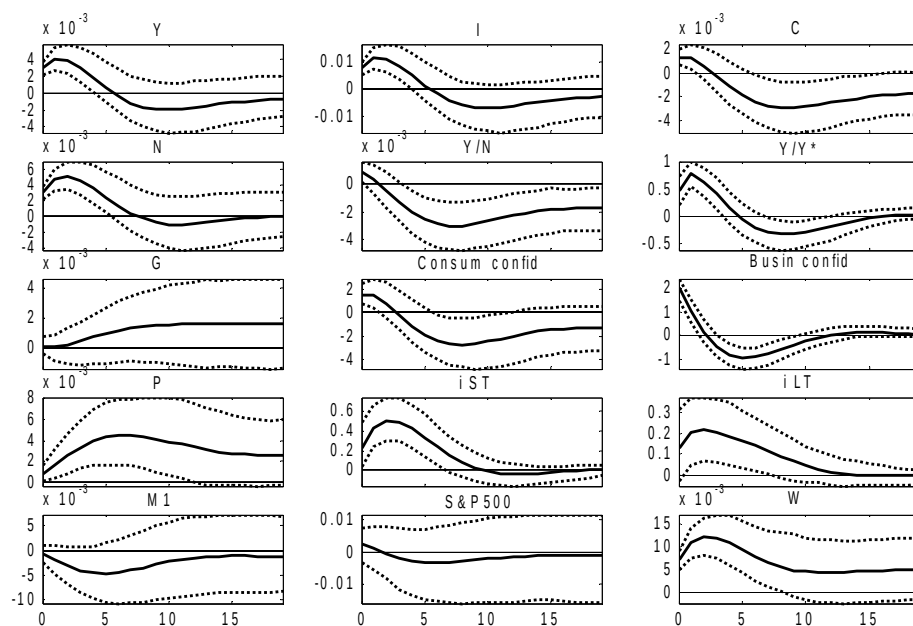
c) US supply shock → US variables



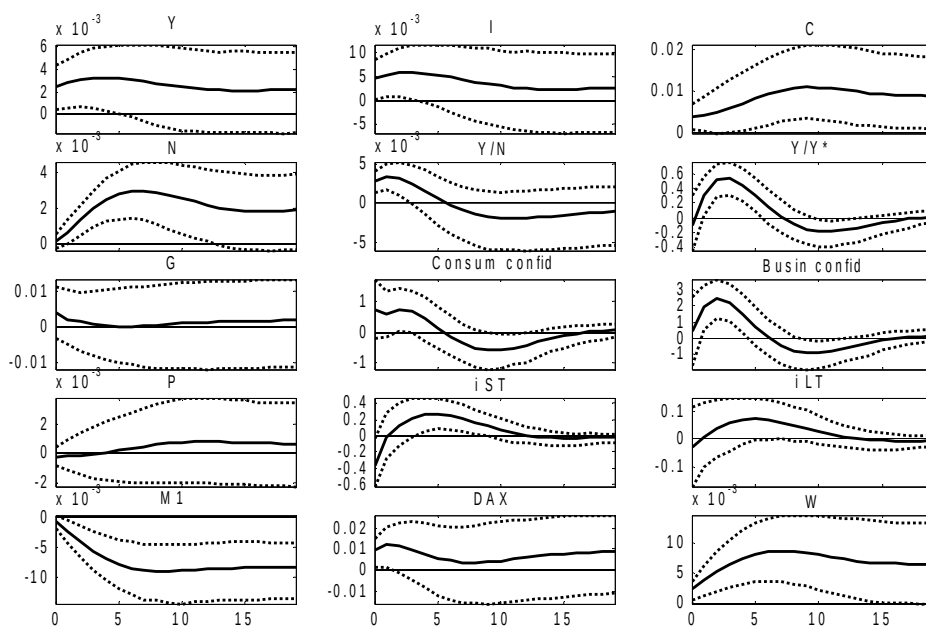
d) US supply shock → German variables



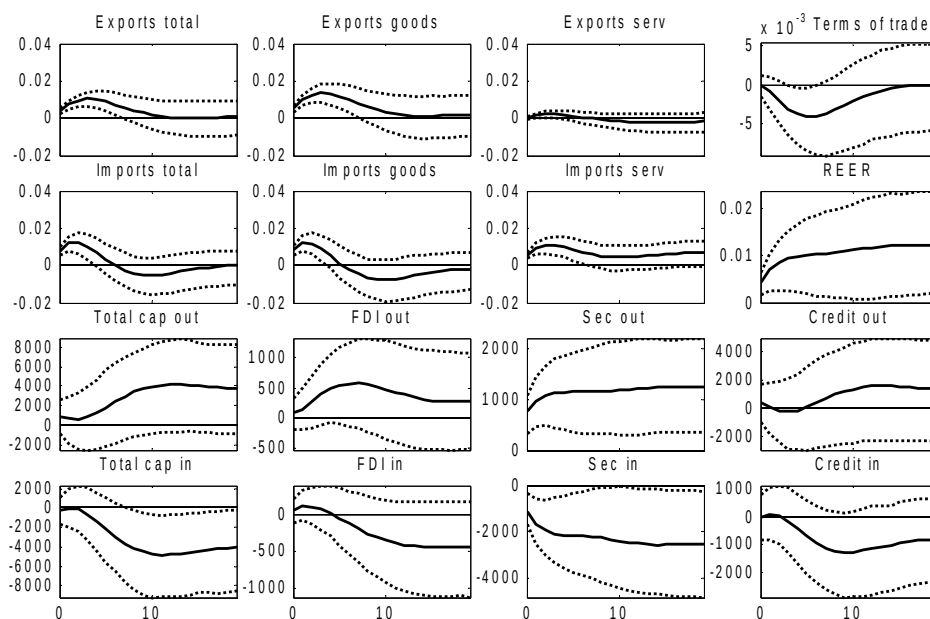
e) US demand shock → US variables



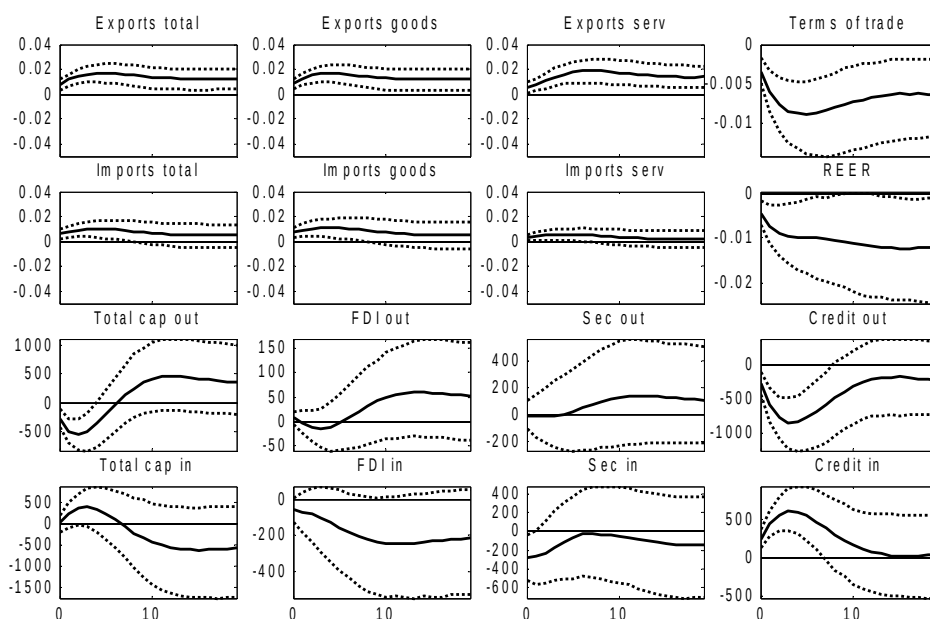
f) US demand shock → German variables



g) US demand shock → US variables



h) US demand shock → German variables



¹⁾ The size of the shocks is one standard deviation. The median and 90% confidence interval are reported here. Y: real GDP. I: real private investment. C: personal consumption expenditure. N: employment. G: real government expenditures. Y/N: labor productivity. Y/Y*: capacity utilization. P: CPI. iST: short-term interest rate. iLT: long-term interest rate. W: wages. REER: real effective exchange rate. Total cap/FDI/Sec/Credit out: net capital outflows. Total cap/FDI/Sec/Credit in: net capital inflows.